Muhammad Bilal Sarwar

+92 3416069355 | bilalsarwar466@gmail.com | www.linkedin.com/in/bilalsarwar071 | Personal Webpage

Research Interests

Formal Verification Methods, Theorem Proving, Higher Order Logic, Model Checking, Probabilistic Analysis

Education

MS in Computational Science & Engineering | ***Advisor: Dr. Osman Hasan*** — Sep. 2022 – Feb. 2025
National University of Sciences & Technology (NUST), Pakistan — CGPA: 3.75/4.0 | Ranked 2nd
Thesis: Formal Analysis of Lane-Changing Algorithms for Autonomous Vehicles Using Probabilistic Model Checking

Bachelor of Science in Electrical Engineering — Oct. 2017 – Oct. 2021
University of Engineering & Technology (UET), Taxila, Pakistan

Publications

Published:

Formal Analysis of Lane-Changing Algorithms using Probabilistic Model Checking — 2026
M.B. Sarwar *and O. Hasan* — J. Signal Process. Syst.

Revolutionizing ICT with AI and ML: A Comprehensive Study of Current Applications and Future Potential — 2024
M.B. Sarwar*, G.M. Raza, M.A. Sarwar, & B.-S. Kim* — IEIE SPC

In Preparation:

Does Reflection Help LLMs Design SAT Heuristics? A Placebo-Controlled Study of Phase-Initialisation Heuristics — 2026
M.B. Sarwar*, O. Hasan and S. Abed* | ***Manuscript completed***

Unified Framework for YouTube Traffic Classification and Out-of-Distribution Detection in Encrypted Networks — 2026
M.B. Sarwar*, S.M. Ahmad, M. Amjad and M. U. S. Khan* | ***To be resubmitted***

Work Experience

Senior Software Engineer | ***GreenAI (Precision Agriculture), HPC Department*** — Feb. 2026 – Present
National Aerospace Science & Technology Park (NASTP) — Rawalpindi, Pakistan

- Built an OpenStack cloud across 20+ bare metal nodes (MAAS, Sunbeam, Juju, MicroCeph, OVN).
- Set up Kubernetes with GPU passthrough and provisioned GPU VMs for internal teams and IaaS clients.
- Deployed vendor-provided and in-house precision agriculture applications from testing to secure production environments, implementing CI/CD pipelines, rollback procedures, and automated nightly backups.
- Lead the HPC Administration team, review configuration changes and prepare the 24/7 shift roster.

Research Assistant | ***System Analysis & Verification (SAVe) Lab*** — Jun. 2025 – Present
SEECS, NUST — Islamabad, Pakistan

- Full-time until Jan. 2026, then part-time and voluntary alongside my role at GreenAI.
- Developing neuro-symbolic optimization frameworks that combine LLMs with SAT-solver heuristics, enabling constraint satisfaction without model fine-tuning.
- Designing and evaluating LLM-guided heuristics for algorithm synthesis, contributing to novel methodologies in automated reasoning and AI-assisted solver design.
- Ran a placebo-controlled study of LLM reflection for SAT heuristics in CaDiCaL. The best evolved heuristics ran 48% (DFVS) and 37% (decision trees) faster than the best default setup.

Research Assistant | ***Data Science & Machine Learning Lab*** — Feb. 2025 – May 2025
SINES, NUST — Islamabad, Pakistan

- Led the lab's project on out-of-distribution (OOD) detection in encrypted YouTube traffic using deep learning.
- Designed the data pipelines and models, and wrote the first draft of the paper, which the lab plans to resubmit.

Graduate Research Student | ***System Analysis & Verification (SAVe) Lab*** Jan. 2024 – Jan. 2025
SEECS, NUST Islamabad, Pakistan

- Verified autonomous vehicle lane-changing algorithms, including MOBIL, with probabilistic model checking.
- Modeled behavior as Markov Decision Processes and verified PCTL properties with **PRISM**.

Research Associate | ***Geo Visual Analytics Lab*** May 2023 – Dec. 2023
IGIS, NUST Islamabad, Pakistan

- Worked on an IoT project to monitor dengue and Zika mosquitoes and find likely outbreak areas.
- Built the real-time data collection setup on Raspberry Pi devices and used OpenCV and machine learning to detect mosquitoes in images.
- Led a 4-week workshop on Python for computer-vision applications.

Technical Support Engineer | ***Technical Department*** Jan. 2022 – Sep. 2022
ZKTeco Pakistan Lahore, Pakistan

- Each week, resolved about 20 Linux and software issues and fixed device connectivity for 15 clients.
- Wrote back-end fixes in Python and Linux and tested each one on the client's system before handover.

Skills

Programming: C/C++, Python, MATLAB, OCaml, Shell Scripting
Formal Methods Tools: HOL4, PRISM
Platforms & Tools: GNU/Linux, Windows/WSL2, Docker, Git, Vim
Cloud & Cluster Management: Slurm, OpenStack, OpenHPC, Kubernetes
ML & DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn
Documentation: LaTeX, Markdown, Microsoft Office Suite

Selected Projects

Multiclass Image Classification using Transfer Learning 2023
Built an image classifier using InceptionResNetV2 with transfer learning, with augmentation and normalization to improve generalization.

Sentiment Analysis Using BERT 2023
Fine-tuned pre-trained BERT models with HuggingFace Transformers for sentiment classification, using NLP preprocessing.

Adaptive Irrigated Agriculture Monitoring (Embedded + Cloud) 2022
Developed an IoT and embedded sensor system for real-time monitoring of soil and air conditions to optimize irrigation.

Selected Honors and Awards

The Punjab Educational Endowment Fund (PEEF), Pakistan 2017–2024
Awarded by the Government of Punjab to exceptional students pursuing higher education. Awarded during my studies at:

- National University of Sciences & Technology (NUST), Pakistan
- University of Engineering & Technology (UET), Taxila, Pakistan

References

Dr. Osman Hasan
Professor, Pro-Rector (Academics), NUST, Islamabad
Reference: Master's Thesis Supervisor
Email: osman.hasan@seecs.nust.edu.pk

Dr. Ammar Mushtaq
Associate Professor, SINES, NUST, Islamabad
Reference: Master's Thesis Co-Supervisor
Email: ammar.mushtaq@sines.nust.edu.pk